



**ДЕПАРТАМЕНТ ОБРАЗОВАНИЯ И НАУКИ
ГОРОДА МОСКВЫ**
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Адаптивные технологии

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Отчётная работа защищена с оценкой «___» _____

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Задание 1

```
SSH: connect to host 10.10.10.10
user@DebianVM:~$ pwd
/home/user
user@DebianVM:~$
```

```
user@DebianVM:~$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILES (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                do not ignore entries starting with .
  -A, --almost-all        do not list implied . and ..
      --author              with -l, print the author of each file
  -b, --escape              print C-style escapes for nongraphic characters
      --block-size=SIZE    with -l, scale sizes by SIZE when printing them;
                           e.g., '--block-size=M'; see SIZE format below
  -B, --ignore-backups     do not list implied entries ending with ~
  -c                        with -lt: sort by, and show, ctime (time of last
                           change of file status information);
                           with -l: show ctime and sort by name;
                           otherwise: sort by ctime, newest first
  -C                        list entries by columns
      --color[=WHEN]       color the output WHEN; more info below
  -d, --directory          list directories themselves, not their contents
  -D, --dired               generate output designed for Emacs' dired mode
  -f                        same as -a -U
  -F, --classify[=WHEN]   append indicator (one of */=>@|) to entries WHEN
                           likewise, except do not append '*'
      --file-type           across, horizontal (-x), commas (-m), long (-l),
                           single-column (-1), verbose (-l), vertical (-C)
      --full-time           like -l --time-style=full-iso
  -g                        like -l, but do not list owner
      --group-directories-first
                           group directories before files
  -G, --no-group            in a long listing, don't print group names
  -h, --human-readable     with -l and -s, print sizes like 1K 234M 2G etc.
      --si                  likewise, but use powers of 1000 not 1024
  -H, --dereference-command-line
                           follow symbolic links listed on the command line
```

```
SSH: 0. Command not found
user@DebianVM:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
user@DebianVM:~$
```

```
user@DebianVM:~$ ls -a
.      .bash_logout  .cache  Desktop  Downloads  .face.icon  .local  Pictures  Public  .sudo_as_admin_successful  Videos
..     .bashrc       .config Documents  .face       .gnupg     Music   .profile  .ssh   Templates
user@DebianVM:~$
```

```
user@DebianVM:~$ cd /bin
user@DebianVM:/bin$
```

```

user@DebianVM:/bin$ ls -l
total 201980
-rwxr-xr-x 1 root root 55720 Jun  4 2025 '['
-rwxr-xr-x 1 root root 38 Oct 13 18:21 7z
-rwxr-xr-x 1 root root 39 Oct 13 18:21 7za
-rwxr-xr-x 1 root root 39 Oct 13 18:21 7zr
-rwxr-xr-x 1 root root 18672 Apr 10 2025 aa-enabled
-rwxr-xr-x 1 root root 18672 Apr 10 2025 aa-exec
-rwxr-xr-x 1 root root 18664 Apr 10 2025 aa-features-abi
-rwxr-xr-x 1 root root 22840 Apr 14 2025 aconnect
lrwxrwxrwx 1 root root 26 Mar  4 2025 addr2line -> x86_64-linux-gnu-addr2line
-rwxr-xr-x 1 root root 183720 May  4 2025 aircan-discover
-rwxr-xr-x 1 root root 47480 Apr 14 2025 alsabat
-rwxr-xr-x 1 root root 81160 Apr 14 2025 alsaloop
-rwxr-xr-x 1 root root 86248 Apr 14 2025 alsamixer
-rwxr-xr-x 1 root root 84280 Apr 14 2025 alsatplg
-rwxr-xr-x 1 root root 31504 Apr 14 2025 alsaucm
-rwxr-xr-x 1 root root 26944 Apr 14 2025 amidi
-rwxr-xr-x 1 root root 59784 Apr 14 2025 amixer
-rwxr-xr-x 1 root root 80232 Apr 14 2025 aplay
-rwxr-xr-x 1 root root 26944 Apr 14 2025 aplaymidi
-rwxr-xr-x 1 root root 22856 Apr 14 2025 aplaymidi2
-rwxr-xr-x 1 root root 14792 Aug 29 2024 appres
-rwxr-xr-x 1 root root 139416 Apr 24 2025 appstreamcli
lrwxrwxrwx 1 root root 6 May  2 2025 apropos -> whatis
-rwxr-xr-x 1 root root 18752 Jun 24 2025 apt
-rwxr-xr-x 1 root root 92552 Jun 24 2025 apt-cache
-rwxr-xr-x 1 root root 27016 Jun 24 2025 apt-cdrom
-rwxr-xr-x 1 root root 31040 Jun 24 2025 apt-config
lrwxrwxrwx 1 root root 29 Jun 24 2025 apt-extracttemplates -> /lib/apt/apt-extracttemplates
-rwxr-xr-x 1 root root 301456 Jun 24 2025 apt-ftparchive
-rwxr-xr-x 1 root root 59784 Jun 24 2025 apt-get
-rwxr-xr-x 1 root root 230 Oct  6 2024 apt-listchanges
-rwxr-xr-x 1 root root 63880 Jun 24 2025 apt-mark
-rwxr-xr-x 1 root root 39240 Jun 24 2025 apt-sortpkgs
lrwxrwxrwx 1 root root 19 Mar  4 2025 ar -> x86_64-linux-gnu-ar
-rwxr-xr-x 1 root root 43464 Jun  4 2025 arch

```

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About

```

user@DebianVM:/bin$ ls -l /sbin
lrwxrwxrwx 1 root root 8 Feb 17 21:26 /sbin -> usr/sbin
user@DebianVM:/bin$

```

```

user@DebianVM:/bin$ ls /sbin -S -h -l -r
lrwxrwxrwx 1 root root 8 Feb 17 21:26 /sbin -> usr/sbin
user@DebianVM:/bin$

```

```

user@DebianVM:/bin$ ls /boot -t -h -l -r
total 128M
-rw-r--r-- 1 root root 12M Dec 30 22:37 vmlinuz-6.12.63+deb13-amd64
-rw-r--r-- 1 root root 83 Dec 30 22:37 System.map-6.12.63+deb13-amd64
-rw-r--r-- 1 root root 277K Dec 30 22:37 config-6.12.63+deb13-amd64
-rw-r--r-- 1 root root 12M Feb  8 10:58 vmlinuz-6.12.69+deb13-amd64
-rw-r--r-- 1 root root 83 Feb  8 10:58 System.map-6.12.69+deb13-amd64
-rw-r--r-- 1 root root 277K Feb  8 10:58 config-6.12.69+deb13-amd64
drwxr-xr-x 5 root root 4.0K Feb 17 22:51 grub
-rw-r--r-- 1 root root 52M Feb 17 22:56 initrd.img-6.12.63+deb13-amd64
-rw-r--r-- 1 root root 53M Feb 17 22:56 initrd.img-6.12.69+deb13-amd64

```

```

user@DebianVM:/bin$ ls /run -X -h -r
crond.reboot tmpfiles.d wpa_supplicant udev sshd setrans network lock gdm3 credentials adduser
gdm3.pid sendsigs.omit.d user systemd speech-dispatcher plymouth mount initramfs dbus console-setup
crond.pid motd.d udisks2 sudo shm NetworkManager log initctl cups avahi-daemon
user@DebianVM:/bin$

```

```

user@DebianVM:/bin$ cd ~
user@DebianVM:~$

```


Задание 2

```
user@DebianVM:~$ cd /tmp
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ ls -d /tmp/newdir
/tmp/newdir
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ rmdir newdir
user@DebianVM:/tmp$ ls
```

```
user@DebianVM:/tmp$ ls -d /tmp/newdir
ls: cannot access '/tmp/newdir': No such file or directory
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ mkdir -p newdir1/newdir2
user@DebianVM:/tmp$ ls -d /tmp/newdir1/newdir2
/tmp/newdir1/newdir2
user@DebianVM:/tmp$
```

```
/tmp/newdir1/newdir2
user@DebianVM:/tmp$ touch newfile1
user@DebianVM:/tmp$ ls -l /tmp/newfile1
-rw-rw-r-- 1 user user 0 Feb 18 01:09 /tmp/newfile1
user@DebianVM:/tmp$
```

```
-rw-rw-r-- 1 user user 0 Feb 18 01:09 /tmp/newfile1
user@DebianVM:/tmp$ >newfile2
user@DebianVM:/tmp$ ls -l /tmp/newfile2
-rw-rw-r-- 1 user user 0 Feb 18 01:11 /tmp/newfile2
user@DebianVM:/tmp$
```

```
-rw-rw-r-- 1 user user 0 Feb 18 01:11 /tmp/newfile2
user@DebianVM:/tmp$ rm newfile2
user@DebianVM:/tmp$ ls /tmp/newfile2
ls: cannot access '/tmp/newfile2': No such file or directory
user@DebianVM:/tmp$
```

Задание 3

```
cp: cannot stat '/etc/fstab': No such file or directory
user@DebianVM:/tmp$ cp /etc/fstab /tmp
user@DebianVM:/tmp$ ls -l /etc/fstab
-rw-r--r-- 1 root root 806 Feb 17 21:26 /etc/fstab
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ ls -l /etc/fstab
-rw-r--r-- 1 root root 806 Feb 17 21:26 /etc/fstab
user@DebianVM:/tmp$
```

```
-rw-r--r-- 1 root root 806 Feb 17 21:26 /etc/fstab
user@DebianVM:/tmp$ cp -p /etc/passwd /tmp
user@DebianVM:/tmp$ ls -l /tmp/passwd
-rw-r--r-- 1 user user 2130 Feb 17 22:51 /tmp/passwd
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ ls -l /tmp/passwd
-rw-r--r-- 1 user user 2130 Feb 17 22:51 /tmp/passwd
user@DebianVM:/tmp$ ls -l /etc/passwd
-rw-r--r-- 1 root root 2130 Feb 17 22:51 /etc/passwd
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ mv fstab fstab_new
user@DebianVM:/tmp$
```

```
bash: mv: command not found
user@DebianVM:/tmp$ ls
fstab_new
```

Задание 4

```
user@DebianVM:/tmp$ mkdir -p dir_a/dir_b
user@DebianVM:/tmp$ ls -ld /tmp/dir_a/dir_b
/tmp/dir_a/dir_b
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ cp -r dir_a dir_a_copy1
user@DebianVM:/tmp$ ls -l /tmp/dir_a_copy1
total 0
drwxrwxr-x 2 user user 40 Feb 19 15:58 dir_b
user@DebianVM:/tmp$ ls -l /tmp/dir_a
total 0
drwxrwxr-x 2 user user 40 Feb 19 15:56 dir_b
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ cp -ra dir_a dir_a_copy2
user@DebianVM:/tmp$ ls -l /tmp/dir_a_copy2
total 0
drwxrwxr-x 2 user user 40 Feb 19 15:56 dir_b
user@DebianVM:/tmp$ ls -l /tmp/dir_a
total 0
drwxrwxr-x 2 user user 40 Feb 19 15:56 dir_b
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ ls dir_a dir_a_new
ls: cannot access 'dir_a': No such file or directory
dir_a_new:
dir_b
```

Задание 5

```
user@DebianVM:/tmp$ touch file1 file2
user@DebianVM:/tmp$ touch file3 file4
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ ls -l file*
-rw-rw-r-- 1 user user 0 Feb 19 16:07 file1
-rw-rw-r-- 1 user user 0 Feb 19 16:07 file2
-rw-rw-r-- 1 user user 0 Feb 19 16:07 file3
-rw-rw-r-- 1 user user 0 Feb 19 16:07 file4

user@DebianVM:/tmp$ rm file*
user@DebianVM:/tmp$ ls -l file*
ls: cannot access 'file*': No such file or directory
user@DebianVM:/tmp$
```

Задание 6

```
user@DebianVM:/tmp$ find ~ -name '.bashrc'
/home/user/.bashrc
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ find ~ -name 'Pictures'
/home/user/Pictures
user@DebianVM:/tmp$
```

Задание 7

```
user@DebianVM:/tmp$ ln -s /etc/fstab fstab_link
user@DebianVM:/tmp$ ls -l fstab_link
lrwxrwxrwx 1 user user 10 Feb 19 16:14 fstab_link -> /etc/fstab
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ ln -s /etc etc_link
```

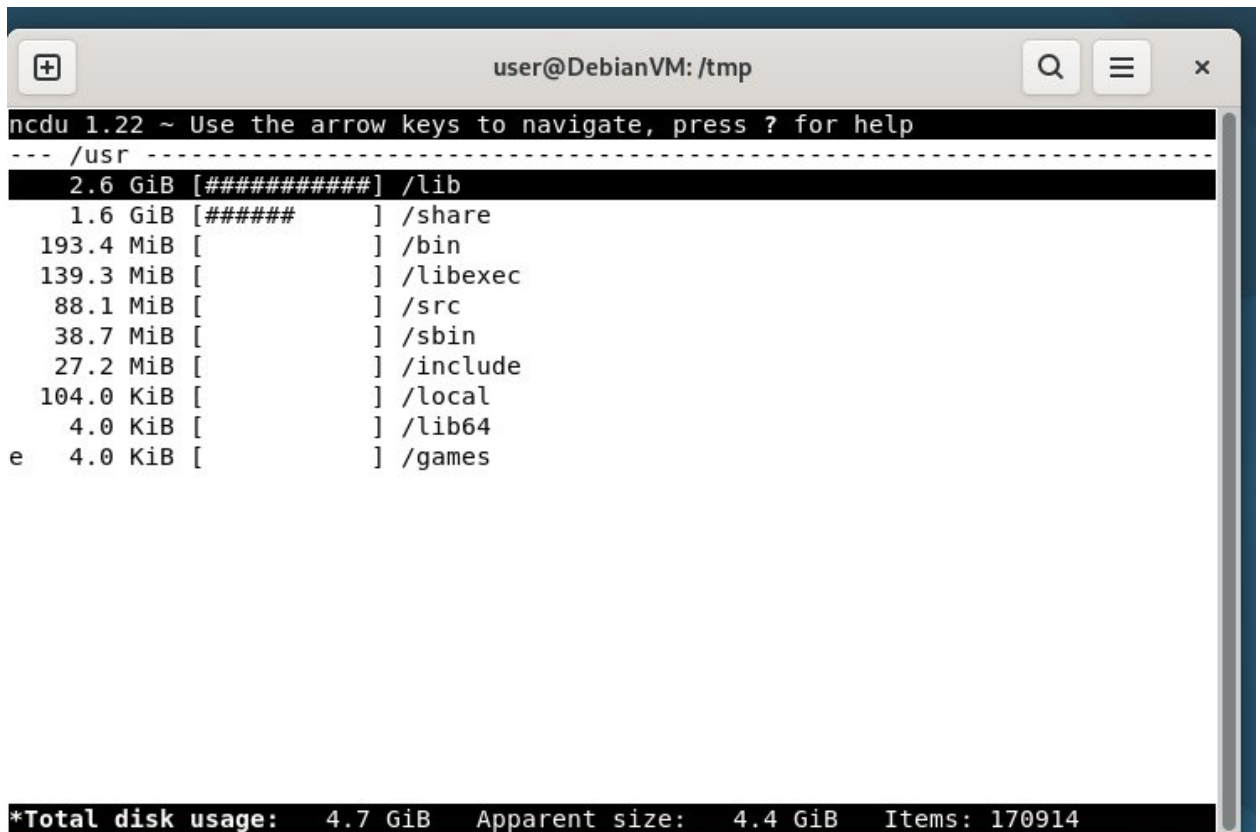
```
user@DebianVM:/tmp$ ls -ld etc_link
lrwxrwxrwx 1 user user 4 Feb 19 16:15 etc_link -> /etc
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ ln -s new_file6 new_file6_link
user@DebianVM:/tmp$ ls -l new_file6_link
lrwxrwxrwx 1 user user 9 Feb 19 16:20 new_file6_link -> new_file6
user@DebianVM:/tmp$
```

```
user@DebianVM:/tmp$ rm new_file6
user@DebianVM:/tmp$ ls -l new_file6_link
lrwxrwxrwx 1 user user 9 Feb 19 16:20 new_file6_link -> new_file6
user@DebianVM:/tmp$
```

Задание 8

```
user@DebianVM:/tmp$ du -s -h /etc
du: cannot read directory '/etc/credstore.encrypted': Permission denied
du: cannot read directory '/etc/ssl/private': Permission denied
du: cannot read directory '/etc/credstore': Permission denied
du: cannot read directory '/etc/chatscripts': Permission denied
du: cannot read directory '/etc/polkit-1/rules.d': Permission denied
du: cannot read directory '/etc/ppp/peers': Permission denied
du: cannot read directory '/etc/cups/ssl': Permission denied
9.8M    /etc
user@DebianVM:/tmp$
```



```
ncdu 1.22 ~ Use the arrow keys to navigate, press ? for help
--- /usr ---
2.6 GiB [#####] /lib
  1.6 GiB [#####] /share
193.4 MiB [ ] /bin
139.3 MiB [ ] /libexec
 88.1 MiB [ ] /src
 38.7 MiB [ ] /sbin
 27.2 MiB [ ] /include
104.0 KiB [ ] /local
   4.0 KiB [ ] /lib64
e  4.0 KiB [ ] /games

*Total disk usage:  4.7 GiB  Apparent size:  4.4 GiB  Items: 170914
```

```
user@DebianVM:/tmp$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G   0    1.9G   0% /dev
tmpfs           393M  1.2M  392M   1% /run
/dev/sda1       19G   5.5G   13G  31% /
tmpfs           2.0G   12K   2.0G   1% /dev/shm
tmpfs           5.0M   8.0K   5.0M   1% /run/lock
tmpfs           1.0M    0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           2.0G   24K   2.0G   1% /tmp
tmpfs           393M  100K  393M   1% /run/user/1000
user@DebianVM:/tmp$
```